Tutorial 2: Group Project

CCST5037
Mathematics: A Cultural Heritage

Learning Outcomes

- By the end of this tutorial you should have

  Formed groups for the group project.

  Understood the requirements for the group project.

  Found some preliminary resources for the group project.

Group Project: Overview

- Group Essay Plan (10% of final grade)
- Group Presentation (15% of final grade)
- Group Essay (20% of final grade)
- **In your project, you should analyze a theme using (at least) one resource (a book or a piece of work) that is relevant to (at least) one of the Course Learning Outcomes (CLOs). The analysis should be based on a hypothesis, and you must argue it using your resource(s) as evidence.**

Our CLOs:

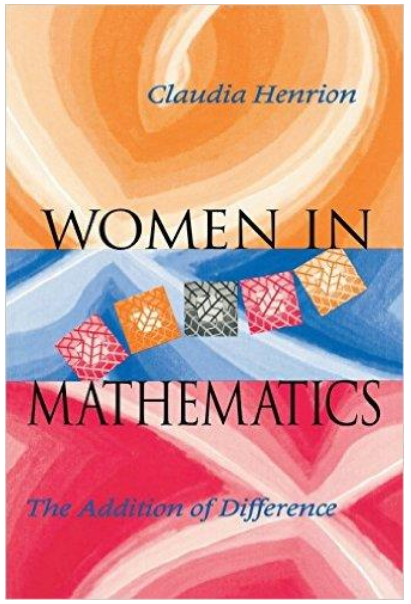
1. Appreciate and describe the beauty, the utility and the "Way" of mathematics.
2. Comprehend and describe how mathematics was and is being developed as a work of human culture.
3. Investigate and describe the interplay among mathematics and other areas of human culture.
4. Investigate and explain the role of mathematics in the development of civilization.

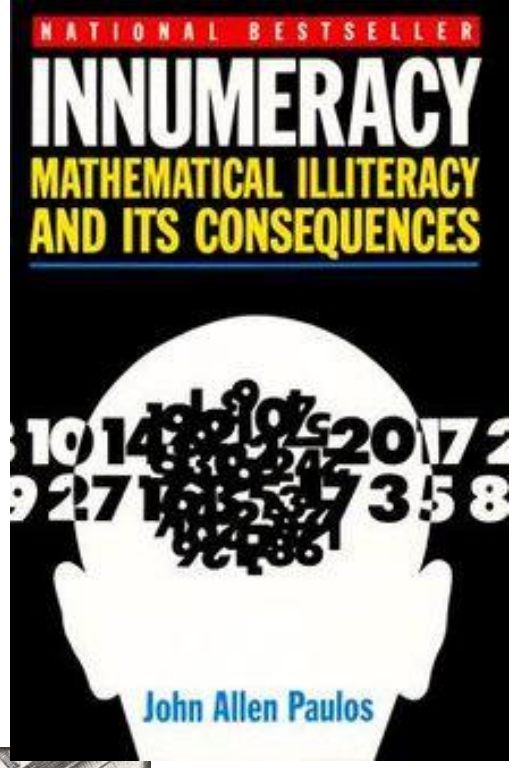
Sample Books/Resources

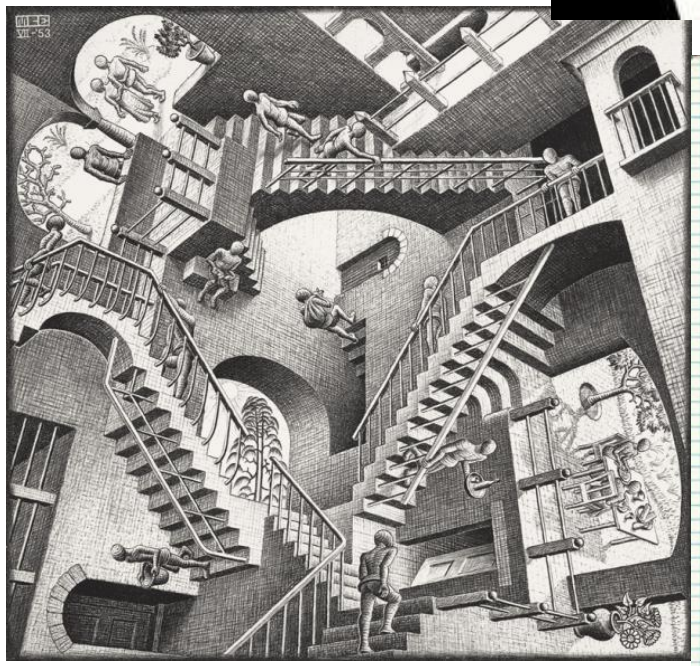
See “Group Project Guidelines.pdf” on Moodle for
a list of possible online resources.

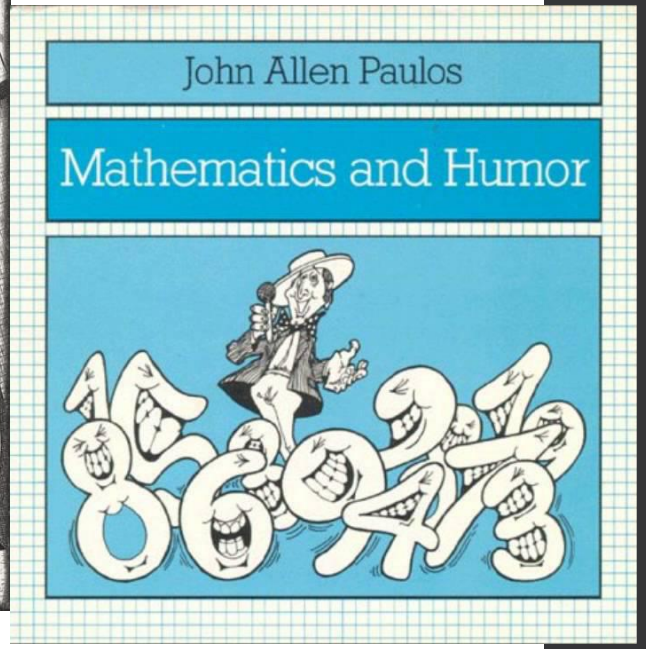
General Interest

- Art of Mathematics
- Women in Mathematics
- Mathematics and Humour
- Topics/Problems of Mathematics
- Math illiteracy
- Cognitive Science
- Biographies



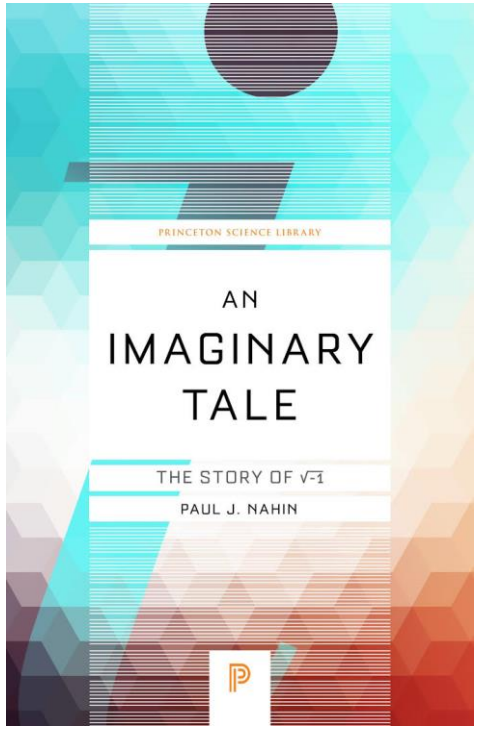


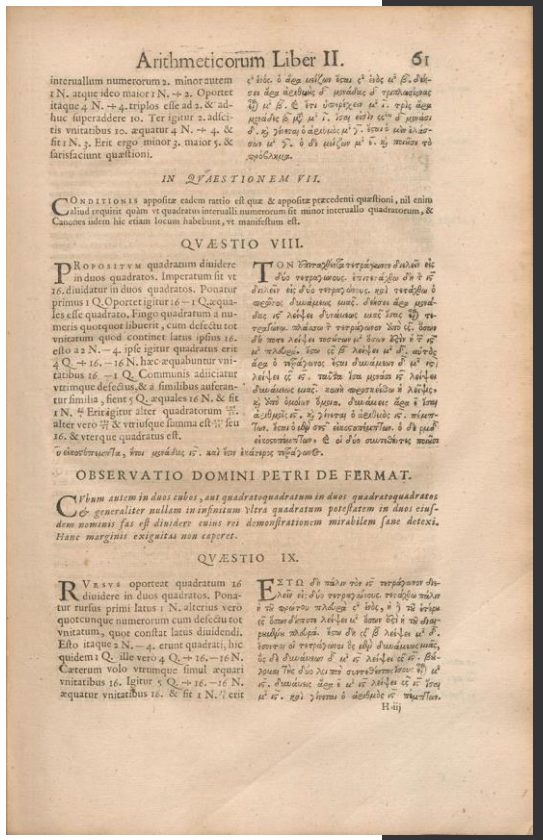


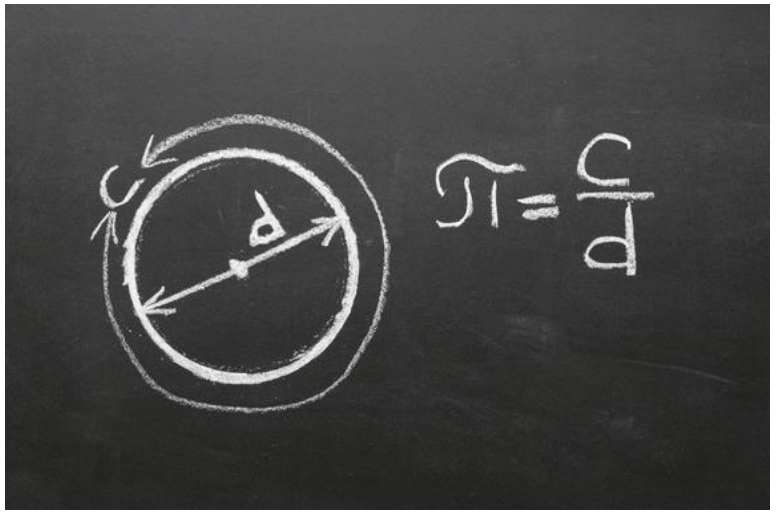


Topics in Mathematics

- $e \approx 2.71818$
  - E. Maor: The Story of a Number
- $\pi \approx 3.1415926536$
  - P. Beckmann, A History of Pi
- $i = \sqrt{-1}$
  - P.J. Nahin, An Imaginary Tale
- Fermat's Last Theorem
  - S. Singh, Fermat's Last Theorem
  - A. Aczel, Fermat's Last Theorem

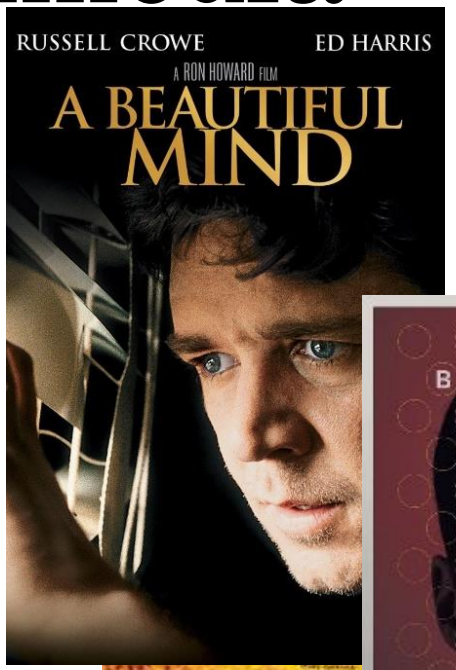


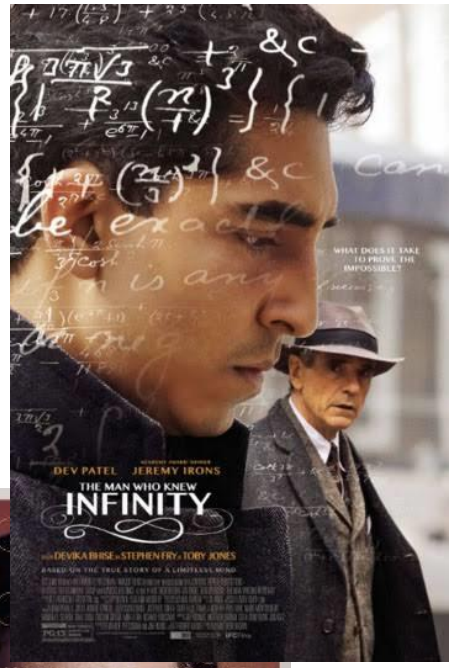




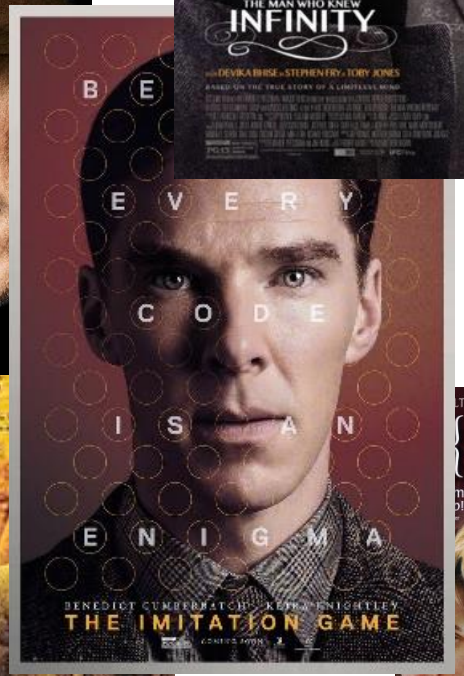
Movies/Other Multimedia

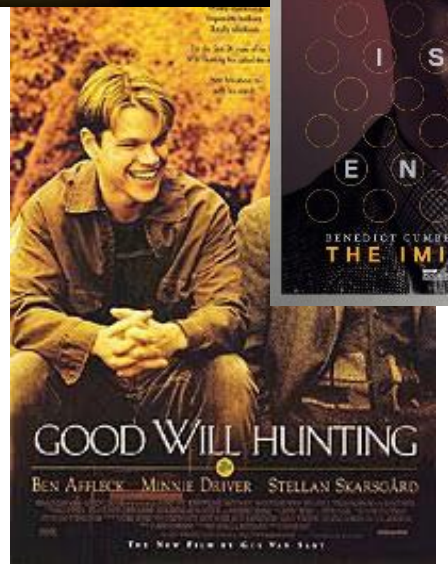
- Good Will Hunting (1997)
- A Beautiful Mind (2001)
- Proof (2005)
- The Housekeeper and the Professor/The Professor and His Beloved Equation (Japanese movie), (2006)
- The Imitation Game (2014)
- The Man Who Knew Infinity (2015)
- Hidden Figures (2017)















Take a minute to think about some topics that interest you in mathematics:

Share them publicly in Zoom Chat.

Form Groups

- Take a few minutes and form your groups.
- All members of a group must be from the same tutorial.
- I will help facilitate if necessary.



Assignment 1: Essay Plan

Due October 16, 16:55

Include in your plan:

i) Tentative project title

ii) Names, UID of group members, and which tutorial group you are in, e.g., "David 04 Tue 16:00"

iii) A list of your primary resource(s) and other references

iv) Brief description (200-400 words) of your essay content, i.e., what is your central theme and how does it relate to the CLOs

- See "Group Project Guidelines.pdf" on Moodle for a list of possible online resources.
- See also "Assignment 1 Description.pdf" for more details and the penalty for late submissions.

Assignment 1: Essay Plan (Response to the feedback)

Due October 29, 16:55

- In addition to receiving a score between 0 and 10 points, you will receive feedback on your assignment.
- In response to this feedback, you can submit an updated essay plan until **16:55, October 29, 2026 (Thursday).** Your assignment 1 score can be increased based on improvements to your plan in the interim.
- Even if you do not choose to submit an updated essay plan, you are strongly encouraged to carefully look at the feedback and take it into account when writing your group essay.
- You won't be penalized even if you don't submit the response to the feedback.

Group Presentation

- **Analyze a theme using (at least) one resource (a book or a piece of work) that is relevant to (at least) one of the Course Learning Outcomes (CLOs). The analysis should be based on a hypothesis, and you must argue it using your resource(s) as evidence.**
- **Dates**: In the last 2 tutorial sessions. The order will be assigned later.
  - Presentation 1: Nov 3/4
  - Presentation 2: Nov 10/11
- **Slides submission deadline:** Sunday Nov 1, 16:55 on Moodle
- **Length:** 12 minutes, + 3 min Q&A.
- **Target audience:** Non-mathematics specific audience.

Additional Points for the Presentation

- You will not be assessed on what you wear, **but you must switch on your webcam during your presentation.**
- **Written/electronic notes** are OK, but remember that you will be assessed on communication skills. This also goes for written notes on your screen (it's easy to tell if you're reading).
- **Videos** (YouTube or otherwise) are OK, but limit the total time to 1 minute. We want to hear YOU speak. Explanations should be given by you.
- You are responsible for testing your hardware/software ahead of time.
- If possible, we recommend just one student controlling the slides.
- **Plagiarism:** Your slides and spoken speech must be your OWN work. Reading verbatim off a website or book is NOT acceptable. You must add your own critical analysis and paraphrase from sources.

Group Essay

- **Length**: 2000-3000 words (excluding bibliography and figures)
- **In your project, you should analyze a theme using (at least) one resource (a book or a piece of work) that is relevant to (at least) one of the Course Learning Outcomes (CLOs). The analysis should be based on a hypothesis, and you must argue it using your resource(s) as evidence.**
- **Essay submission deadline:** December 1, 16:55 (Tuesday)
- Late submissions will not be graded.
- See "Group Project Guidelines.pdf" on Moodle for a list of possible online resources.



Discuss possible topics within your group:

If you want resources not on the list, come check with me.

Breakout Rooms, I'll come visit.

Summary

- For those of you who asked me about references, I'll reply by e-mail.
- Debate next tutorial: **Should we memorize times (multiplication) tables or other mathematical facts?**
  - Prepare both sides, we'll choose sides for you next week. Read both articles in "Tutorials" for Debate 1.
    - Search for your own articles too.
  - Counts towards your tutorial participation marks.
- E-mail us with any questions or concerns:
  - Toby tobylai@hku.hk
  - David davidyu@hku.hk